SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

Wireless Communication & Networking (EGE321)

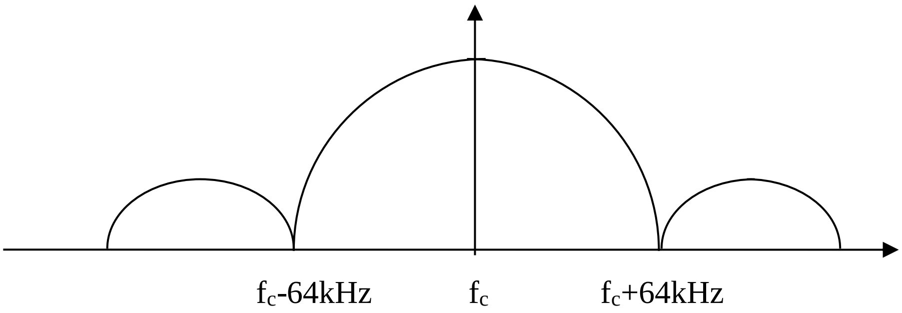
Tutorial 7: Digital Modulation

1) Sketch the constellation diagram of the BPSK modulation. Explain how information is represented in BPSK modulation.

2) Is it possible for the BPSK signal to have zero RF envelope? Explain your answer.

3) Calculate and draw the mainlobe bandwidth of an RF signal modulated by a 2Mbps random data sequence onto a 900MHz carrier, assume no baseband filtering is used and the modulation scheme is QPSK. What would be the RF mainlobe bandwidth if 8PSK is used?

4) Figure below shows the spectral occupancy of a typical MPSK signal modulated on a carrier frequency of 900kHz. If the bit rate is 256kbps, what is the value of M?



5) In order to limit the bandwidth, a Raised Cosine filter with roll off factor, $\alpha = 0.35$ is used, draw the RF bandwidth at question 2.

6) State the difference between Binary Signaling and M-ary Signaling.

7) QPSK modulation is widely used in digital communication systems. Sketch the Vector diagram for QPSK modulation and list the possible phase shifts in QPSK modulation.